B.Sc. (Entire Computer Science)-I, Level - 4.5 UG Certificate Level					
Sem:	II				
Paper Category:	DSC2-2 **(Major)**				
Paper Name:	Python Programming -II	**Paper Code :** G06-0202			
Credit:	02	**Theory:**	2 Hrs./Week		
Marks:	**UA:** 30	**CA:** 20	**Total:** 50		

Course Objective:

1. To learn the use of functions in programming.
2. To understand the use of modules and packages in the application hierarchy.
3. To understand Python programming using object-oriented programming principles.
4. To learn handling of various exceptions during the application development.
5. To understand the working with different file operations.

Course Outcomes: Upon successful completion of this course, students will be able to

1. Write and implement a functional approach to application development.
2. Write and implement a modular approach to application development.
3. Design an application using an object-oriented paradigm.
4. Create error-free applications by applying the exception-handling concept.
5. Design an application that contains the use of different files for data processing.

Unit I: Modules and packages	No. of Lectures:15	Weightage: 8-10 Marks

Introduction: Introduction to modules and packages, import statement, from…import statement, creating our own modules, working with built-in modules- Math module, time module and random module.

Python Object Oriented: Difference between procedure-oriented and object-oriented programming. Features of object-oriented programming- classes and objects, inheritance, polymorphism, encapsulation, abstraction. Creating class, self-variable, constructor, types of variables, namespaces, types of methods, passing member of one class to another class, inner classes. Types of inheritance, super() method, method overloading, method overriding, operator overloading, abstract classes, and interfaces.

Unit II: Exception Handling and Threading:	No. of Lectures:15	Weightage: 20-22 Marks

Threading: Understanding threads, Class and threads, Creating Threads, Thread Synchronization, Treads Life cycle, multi-threading.

Exception Handling: Error in Python program, exceptions, steps in exception handling using try, except, else and finally blocks, types of exceptions- built-in and user-defined exceptions, assert statement.

File Input Output: Types of files in Python, opening a file- the file opening modes, closing a file, working with text files containing strings, working with binary files, with statement, pickling and unpickling, seek() and tell() methods, random accessing of binary files, zipping and unzipping files, working with directories.

Reference Books:

1. Python: The Complete Reference by Martin C. Brown.
2. Core Python Programming, Dreamtech publications, by R. Nageswara Rao.
3. Python Programming, A modular approach, First Edition, Pearson, by Taneja Sheetal
4. Learning with Python, Dreamtech publications, by Allen Downey
5. Python Programming for the Absolute Beginner by Michael Dawson-Cengage Learning.